

Math Analysis

Ryan Vaz

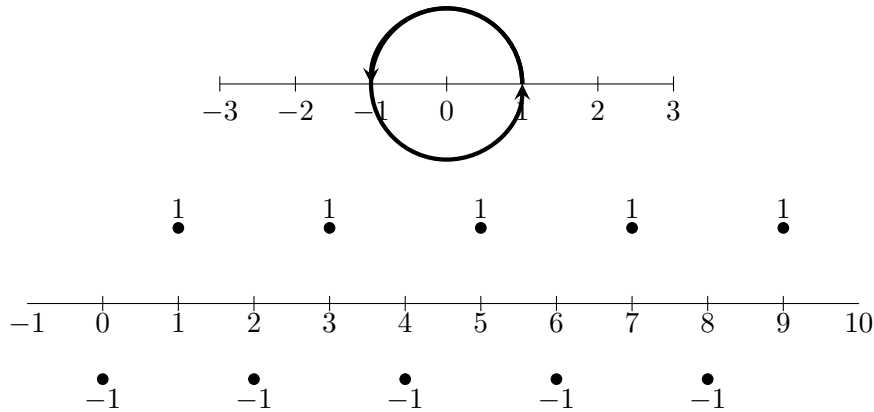
1 The Real Numbers

1.1 Sequences and Limits

Definition 1.1. Sequences A sequence of real numbers exists if we have a map $a : \mathbb{N} \rightarrow \mathbb{R}$ denoted $(a_1, a_2, a_3, \dots)$ which is an infinite list. This is denoted as $(a_n)_{n \in \mathbb{N}}$ or $(a_n)_{n=1}^{\infty}$ or (a_n) if the context is clear.

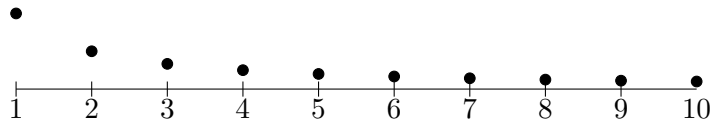
Example 1-

$$(a_n)_{n \in \mathbb{N}} = ((-1)^n)_{n \in \mathbb{N}} = (-1, 1, -1, 1, \dots)$$



Example 2-

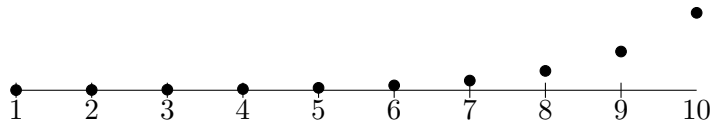
$$(a_n)_{n \in \mathbb{N}} = \left(\frac{1}{n}\right)_{n \in \mathbb{N}} = \left(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right)$$



$$\lim_{n \rightarrow \infty} a_n = 0$$

Example 3-

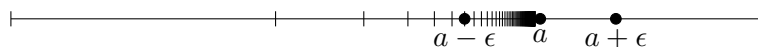
$$(a_n)_{n \in \mathbb{N}} = (2^n)_{n \in \mathbb{N}} = (2, 4, 8, 16, 32, 64, 128, 256, \dots)$$



$$\lim_{n \rightarrow \infty} a_n = \infty$$

Definition 1.2. Convergent Sequences A sequence is said to be convergent to $a \in \mathbb{R}$ if a_n lies in the epsilon-neighborhood of a or $\forall \epsilon > 0, \exists N \in \mathbb{N}$ where $\forall n \geq N : |a_n - a| < \epsilon$. This condition must be satisfied no matter how small epsilon gets. Eventually all the sequence elements must lie within any given epsilon neighborhood in order for it to converge (infinitely many inside and finitely many outside).

Example of a sequence that converges within a's ϵ neighborhood



Definition 1.3. Divergent Sequences A sequence is said to be divergent if there is no such limit point $a \in \mathbb{R}$.

Prove $(a_n)_{n \in \mathbb{N}} = (1/n)_{n \in \mathbb{N}}$ is convergent to $0 \in \mathbb{R}$.

In these types of proofs, we always will use the same first and last arguments. First is that let $\epsilon > 0$. Last is that the conditional $|a_n - 0| < \epsilon$ will hold for any $n \geq N$ (any n value past N is going to be part of the ϵ neighborhood). From here it's our job to fill in the middle steps.

We know $|a_n - 0| < \epsilon$ must hold, now we can perform calculations on it.

$$\begin{aligned} |a_n - 0| &< \epsilon \\ |a_n| &< \epsilon \\ (a_n)_{n \in \mathbb{N}} &= (1/n)_{n \in \mathbb{N}} \\ 1/n &< \epsilon \\ n &\geq N \\ 1/n &\leq 1/N \\ 1/n &\leq 1/N < \epsilon \end{aligned}$$

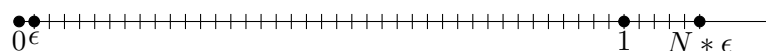
Now we can choose any value for N , so let's make life easy and choose an N so big that $N * \epsilon > 1$. This allows us to state the whole proof as follows:

Let $\epsilon > 0$. We choose $N \in \mathbb{N}$ s.t. $N * \epsilon > 1$. $\forall n \geq N$, we have $|a_n - 0| = |a_n| = 1/n \leq 1/N \leq \epsilon$

Nothing in life is easy though... who said we can assume that?

Definition 1.4. Archimedean Property No matter how small a number(ϵ) is, we can always add a finite amount of itself to be greater than(not necessarily equal) to any given number.

Archimedean Property visualized



Prove $(a_n)_{n \in \mathbb{N}} = ((-1)^n)_{n \in \mathbb{N}}$ is divergent.

For this type of problem, we can use the exact same approach as the previous problem, but this time using a proof by contradiction.

Let's assume the sequence converges to $a \in \mathbb{R}$. By intuition we can see that this sequence doesn't converge to a single value a (see example 1) but sadly fancy diagrams just won't cut it here, we need to prove it!

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \forall n \geq N : |a_n - a| < \epsilon$$

We know the distance between -1 and 1 is 2 , thus ϵ must be lower than that, so let's choose $\epsilon = 1$. This means $|a_N - a| < \epsilon$ for any terms at N and past N .

We can extend this to say:

$$\begin{aligned} |a_N - a| &< \epsilon \\ |a_{N+1} - a| &< \epsilon \end{aligned}$$

and we know that every two terms are going to be either -1 and 1 or 1 and -1, thus

$$|1 - a| < \epsilon \text{ and } |(-1) - a| < \epsilon$$

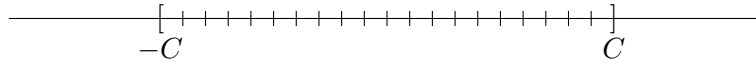
therefore

$$2 = |1 - (-1)| = |1 - a + a - (-1)| \leq |1 - a| + |a - (-1)| = |1 - a| + |(-1) - a| < 2$$

We have achieved a contradiction $2 < 2$!

Definition 1.5. Boundedness A sequence $(a_n)_{n \in \mathbb{N}}$ is called bounded if $\exists C \in \mathbb{R}$ s.t. $\forall n \in \mathbb{N} : |a_n| \leq C$. Otherwise, the sequence is unbounded.

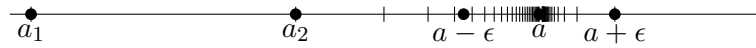
Entire sequence is bounded by C



Observation 1.6. $(a_n)_{n \in \mathbb{N}}$ convergent $\implies (a_n)_{n \in \mathbb{N}}$ bounded
 $(a_n)_{n \in \mathbb{N}}$ unbounded $\implies (a_n)_{n \in \mathbb{N}}$ does not converge

Proof Given a sequence $(a_n)_{n \in \mathbb{N}}$, if it converges to some value a , it has some points around its ϵ -neighborhood. By this logic we can define some value C to be the member that is furthest away from the ϵ -neighborhood. Thus we can say $C := \max(|a_1|, |a_2|, |a_3|, \dots, |a_{n-1}|, |a + \epsilon|)$.

Sequence bounded by a_1



Observation 1.7. $(a_n)_{n \in \mathbb{N}}$ convergent $\implies$ There is only one limit

Proof Assume we have two distinct limits, a and $\tilde{a}$.

We know $|a - \tilde{a}| > 0$, let's say $\epsilon := \frac{1}{4}|a - \tilde{a}| > 0$.

Then: $\exists N \in \mathbb{N} \forall n \geq N : |a_n - a| < \epsilon$

Also: $\exists \tilde{N} \in \mathbb{N} \forall n \geq \tilde{N} : |a_n - \tilde{a}| < \epsilon$

Thus: For $n \geq \max(N, \tilde{N}) : |a - \tilde{a}| = |a - a_n + a_n - \tilde{a}| \leq |a - a_n| + |a_n - \tilde{a}|$ by the triangle inequality

Substituting the above for ϵ gives us

$$|a - \tilde{a}| < 2\epsilon$$

Using our original definition for ϵ leads to

$$|a - \tilde{a}| < \frac{1}{2}|a - \tilde{a}|$$

Which is our contradiction, thus a convergent sequence has a unique limit.

1.2 Limit Theorems

We express limits normally using

$$\lim_{n \rightarrow \infty} a_n = a$$

However there's alternate ways we can express them, for example

$$a_n \xrightarrow{n \rightarrow \infty} a$$

or

$$a_n \xrightarrow{n \rightarrow \infty} a$$

For any given convergent sequences $(a_n)_{n \in \mathbb{N}}$ and $(b_n)_{n \in \mathbb{N}}$:

1. Given that a and b are convergent

$$\lim_{n \rightarrow \infty} a_n + b_n = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n$$

2. Given that a and b are convergent

$$\lim_{n \rightarrow \infty} a_n * b_n = \lim_{n \rightarrow \infty} a_n * \lim_{n \rightarrow \infty} b_n$$

3. Given that a and b are convergent and the denominator is non zero

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}$$

Proof Is the following a convergent sequence, if so what is it's limit?

$$\frac{2n^2 + 5n - 1}{-5n^2 + n + 1}$$

We know the most important term is n^2 so we just need to expand it with the highest power

$$\frac{\frac{1}{n^2} 2n^2 + 5n - 1}{\frac{1}{n^2} - 5n^2 + n + 1} = \frac{2 + 5/n - 1/n^2}{-5 + 1/n + 1/n^2}$$

We proved $\frac{1}{n} \xrightarrow{n \rightarrow \infty} 0$ earlier, and by the second limit theorem $\frac{1}{n} * \frac{1}{n} \xrightarrow{n \rightarrow \infty} 0$ thus we can say

$$\frac{2 + 5/n - 1/n^2}{-5 + 1/n + 1/n^2} \xrightarrow[n \rightarrow \infty]{\text{Limit Theorems}} \frac{2 + 0 - 0}{-5 + 0 + 0} = \frac{-2}{5}$$

For any given convergent sequences $(a_n)_{n \in \mathbb{N}}$ and $(b_n)_{n \in \mathbb{N}}$:

1. Monotonicity:

$$a_n \leq b_n \forall n \in \mathbb{N} \implies \lim_{n \rightarrow \infty} a_n \leq \lim_{n \rightarrow \infty} b_n$$

2. Sandwich theorem:

$$\begin{aligned} a_n \leq c_n \leq b_n \forall n \in \mathbb{N} \text{ and } \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n \\ \implies (c_n)_{n \in \mathbb{N}} \text{ convergent with } \lim_{n \rightarrow \infty} c_n = \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n \end{aligned}$$

Proof By limit theorems,

$$\begin{aligned} (b_n - a_n) \xrightarrow[n \rightarrow \infty]{n \rightarrow \infty} \lim_{n \rightarrow \infty} b_n - \lim_{n \rightarrow \infty} a_n = 0 \\ d_n := c_n - a_n \implies 0 \leq d_n \leq b_n - a_n \end{aligned}$$

Let $\epsilon > 0$. $\exists N \in \mathbb{N}$ s.t. $\forall n \geq N : |b_n - a_n| < \epsilon$ thus $|d_n - 0| \leq |b_n - a_n| < \epsilon$

$\implies (d_n)_{n \in \mathbb{N}}$ converges with limit 0

$\implies (c_n)_{n \in \mathbb{N}} = (d_n + a_n)_{n \in \mathbb{N}}$ is convergent with limit a

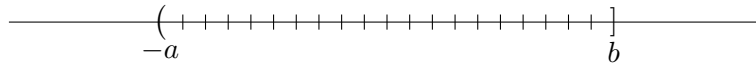
Proof Is the sequence $(c_n)_{n \in \mathbb{N}}$ given by $c_n = \sqrt{n^2 + 1} - n$ convergent, if so what is the limit?

$$\begin{aligned} c_n &= \sqrt{n^2 + 1} - n \\ &= (\sqrt{n^2 + 1} - n) * \frac{(\sqrt{n^2 + 1} + n)}{(\sqrt{n^2 + 1} + n)} \\ &= \frac{n^2 + 1 - n^2}{(\sqrt{n^2 + 1} + n)} \\ &= \frac{1}{(\sqrt{n^2 + 1} + n)} \end{aligned}$$

$$\sqrt{n^2 + 1} > 0 \forall n \in \mathbb{N} \implies \frac{1}{(\sqrt{n^2 + 1} + n)} \leq \frac{1}{n} \implies 0 \leq c_n \leq \frac{1}{n} \forall n \in \mathbb{N} \implies \lim_{n \rightarrow \infty} c_n = 0$$

1.3 Supremum and Infimum

Sequence bounded by a and b



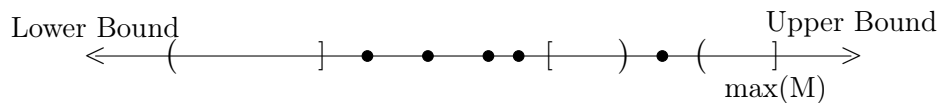
We denote this interval as $(a, b] := \{x \in \mathbb{R} | a < x \leq b\}$

If we wanted to include a as a part of the interval we'd use $[a, b] := \{x \in \mathbb{R} | a \leq x \leq b\}$

And if we don't have a bound on one side, we denote that as $[a, \infty) := \{x \in \mathbb{R} | a \leq x\}$

These are very nice subsets of the real number line, but sometimes it can be nasty, for example

Sequence M



Recall that we define a bound to be a term for which no term in the sequence exceeds it, aka $\nexists a \in M$ s.t. $|a| > |C|$ where C is a bound.

We can split this definition into two terms, the upper and lower bounds.

Upper Bound A term(not necessarily in the sequence) that is greater than or equal to all the terms in the sequence.

$$b \in \mathbb{R} \text{ is an upper bound for } M \text{ if } \forall x \in M : x \leq b$$

Lower Bound A term(not necessarily in the sequence) that is less than or equal to all the terms in the sequence.

$$a \in \mathbb{R} \text{ is a lower bound for } M \text{ if } \forall x \in M : x \geq a$$

There can exist multiple upper and lower bounds in a sequence, for example any point above and including $\max(M)$ will be an upper bound.

If b is an upper bound of M and $b \in M$, then b is called a maximal element of M (and vice versa for minimal element)

Now we run into a problem...what exactly is the $\min(M)$? We put it at a point above, but notice how our set does not include M(think hollow dot on a number line). Therefore we can say $\min(M)$ does not exist. This will lead us to our next definitions.

Suppose we have the following interval

$M = (1, 3)$, $\max(M)$, $\min(M)$ do not exist

Sequence M bounded by 1 and 3



We can choose upper bound to be any term such as 5, but let's try to choose a number closer to M, this will lead us to 4, and eventually 3 as the lowest upper bound. This is our definition for supremum, the least upper bound.

Definition 1.8. Supremum For a subset $M \subset \mathbb{R} : s \in \mathbb{R}$ is called a supremum of M if $\forall x \in M : x \leq s$ (upper bound for M) and $\forall \epsilon > 0 \exists \tilde{x} \in M : s - \epsilon < \tilde{x}$ ($s - \epsilon$ is no longer an upper bound for M) which will be denoted as $\sup(M) := s$ (or $\sup(M) := \infty$ if M is not bounded above)

Definition 1.9. Infimum For a subset $M \subset \mathbb{R} : l \in \mathbb{R}$ is called a supremum of M if $\forall x \in M : x \geq l$ (lower bound for M) and $\forall \epsilon > 0 \exists \tilde{x} \in M : l + \epsilon > \tilde{x}$ ($l + \epsilon$ is no longer an lower bound for M) which will be denoted as $\inf(M) := l$ (or $\inf(M) := -\infty$ if M is not bounded below)

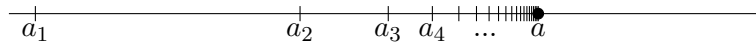
Now you may be wondering about the empty set, well by a vacuous truth all terms of the empty set are upper bounds therefore $\sup(\emptyset) = -\infty$ and likewise, all the terms of the empty set are lower bounds therefore $\inf(\emptyset) = \infty$

1.4 Cauchy sequences

Remember our old definition for convergence? We have an issue with it, namely that nasty limit. We can't always solve it, so we need an alternative way to deal with it.

Let's present a new idea:

Suppose $(a_n)_{n \in \mathbb{N}}$ is convergent with $a_n \xrightarrow{n \rightarrow \infty} a$



We don't want to use a to describe this limit though, but we can use the terms of the sequence. We can say that the terms of the sequence will lie arbitrarily close to each other eventually.

Definition 1.10. Cauchy Sequence If $\forall \epsilon > 0 \exists N \in \mathbb{N}$ s.t. $\forall n, m \leq N : |a_n - a_m| < \epsilon$ then $(a_n)_{n \in \mathbb{N}}$ is called a Cauchy sequence

Observation 1.11. Cauchy Sequence $\iff$ Convergent Sequence

Definition 1.12. Dedekind Completeness

If $M \subset \mathbb{R}$ is bounded above, then $\sup(M) \in \mathbb{R}$ (exists)

If $M \subset \mathbb{R}$ is bounded below, then $\inf(M) \in \mathbb{R}$ (exists)

Observation 1.13. If $(a_n)_{n \in \mathbb{N}}$ is monotonically decreasing ($a_{n+1} \leq a_n \forall n$) and bounded below (the set $(a_n)_{n \in \mathbb{N}}$ has a lower bound), then: $(a_n)_{n \in \mathbb{N}}$ is convergent and

If $(a_n)_{n \in \mathbb{N}}$ is monotonically increasing ($a_{n+1} \geq a_n \forall n$) and bounded above (the set $(a_n)_{n \in \mathbb{N}}$ has an upper bound), then: $(a_n)_{n \in \mathbb{N}}$ is convergent

Definition 1.14. Subsequences Let $(n_k)_{k \in \mathbb{N}}$ be a sequence of natural numbers that is strictly monotonically increasing, then $(a_{n_k})_{k \in \mathbb{N}}$ is called a subsequence of $(a_n)_{n \in \mathbb{N}}$. $\forall k \in \mathbb{N} : n_{k+1} > n_k$

Example: $(a_n)_{n \in \mathbb{N}}$ given by $a_n = 1/n$, $(1, 1/2, 1/3, 1/4, 1/5, 1/6, 1/7, 1/8)$

Let $(a_{n_k})_{k \in \mathbb{N}} = (a_{2^k})_{k \in \mathbb{N}} = (1/2, 1/4, 1/8, 1/16, \dots)$

Observation 1.15. $(a_n)_{n \in \mathbb{N}}$ convergent with $a_n \xrightarrow{n \rightarrow \infty} a \implies$ every subsequence $(a_{n_k})_{k \in \mathbb{N}}$ is convergent with $a_{n_k} \xrightarrow{n \rightarrow \infty} a$

Definition 1.16. Accumulation Values $a \in \mathbb{R}$ is called an accumulation value of $(a_n)_{n \in \mathbb{N}}$ if there is a subsequence $(a_{n_k})_{k \in \mathbb{N}}$ with $a_{n_k} \xrightarrow{k \rightarrow \infty} a$. Also called a cluster point, accumulation point, limit point, or partial limit.

$a \in \mathbb{R}$ is an accumulation value of $(a_n)_{n \in \mathbb{N}}$ $\iff \forall \epsilon > 0 : \text{The } \epsilon\text{-neighbourhood of } a \text{ contains infinitely many sequence members of } (a_n)_{n \in \mathbb{N}}$

Observation 1.17. Convergent Sequences have one accumulation value

Divergent Sequences have multiple accumulation values

Definition 1.18. Bolzano-Weierstrass Theorem $(a_n)_{n \in \mathbb{N}}$ bounded $\implies (a_n)_{n \in \mathbb{N}}$ has an accumulation value aka has a convergent subsequence.

Proof: Suppose we have an interval bounded by $[c_0, d_0]$ and split it in half.

We can split the interval in half, and pick the one with infinitely many terms in it and ignore the rest. This gives us an interval that is half the length of the previous one so either the Upper or Lower bounds changed, so we update the names $[c_1, d_1]$.

Now we repeat the step and bisect the interval to form $[c_n, d_n]$ which is a proper subset of our original interval thus we have...

$$[c_0, d_0] \supset [c_1, d_1] \supset [c_2, d_2] \supset \dots$$

and

$$d_1 - c_1 = 1/2(d_0 - c_0), d_2 - c_2 = 1/2(d_1 - c_1) = 1/4(d_0 - c_0)$$

thus

$$d_n - c_n = 1/2^n(d_0 - c_0) \xrightarrow{n \rightarrow \infty} 0$$

We know:

$(c_n)_{n \in \mathbb{N}}$ monotonically increasing and bounded $(d_n)_{n \in \mathbb{N}}$ monotonically decreasing and bounded $\implies$

$(c_n)_{n \in \mathbb{N}}, (d_n)_{n \in \mathbb{N}}$ are convergent

By limit theorems: $0 = \lim_{n \rightarrow \infty} (d_n - c_n) = \lim_{n \rightarrow \infty} d_n - \lim_{n \rightarrow \infty} c_n$

Define a subsequence $(a_{n_k})_{k \in \mathbb{N}}$ by choosing $a_{n_k} \in [c_k, d_k] \implies c_k \leq a_{n_k} \leq d_k \implies (a_{n_k})_{k \in \mathbb{N}}$ is convergent (by sandwich theorem)

Definition 1.19. Limit superior/inferior Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers. An element $a \in \mathbb{R} \cup \{-\infty, \infty\}$ is a:

- Limit superior of $(a_n)_{n \in \mathbb{N}}$ if a is the largest (improper) accumulation value of $(a_n)_{n \in \mathbb{N}}$
Write:

$$a = \limsup_{n \rightarrow \infty} a_n$$

- Limit inferior of $(a_n)_{n \in \mathbb{N}}$ if a is the smallest (improper) accumulation value of $(a_n)_{n \in \mathbb{N}}$
Write:

$$a = \liminf_{n \rightarrow \infty} a_n$$

2 The Topology of the Real Number Line

For any point on a number line, we have a epsilon neighborhood around it, i.e. For $\epsilon > 0$: $(x - \epsilon, x + \epsilon) =: B_\epsilon(x)$ which represents the maximal distance from x .

If we don't need to say how close it is, instead of saying an ϵ -neighborhood

$M \subset \mathbb{R}$ is called a neighbourhood of x if there is $\epsilon > 0$ s.t. $M \supset B_\epsilon(x)$

For example,

$[-2, 2]$ is a neighborhood of 0

$[-2, 2]$ is a neighborhood of 1

This is because if we can find an ϵ s.t. $\epsilon > 0$ and interval $[a \pm \epsilon] \subset S$ then it is a valid neighborhood.

We cannot find such a neighborhood for a value such as 2 because there is no ϵ s.t. $2 + \epsilon \in [-2, 2]$

Definition 2.1. Open Set A set $M \subset \mathbb{R}$ is called open if, $\forall x \in M, M$ is a neighborhood of x .
 $\forall x \in M \exists \epsilon > 0 : M \supset B_\epsilon(x)$

Definition 2.2. Closed Set $A \subset \mathbb{R}$ is called closed (in $\mathbb{R}$) if $A^c := \mathbb{R} \setminus A$ is open

Example:

$\emptyset, \mathbb{R}$ are both open and closed

$[-2, 2]$ is closed but not open

$(-2, 2)$ is open but not closed

$(-2, 2]$ is neither open or closed

Observation 2.3. $A \subset \mathbb{R} \iff \forall$ convergent sequences $(a_n)_{n \in \mathbb{N}}$ with $a_n \in A \forall n \in \mathbb{N}, a_n \xrightarrow{n \rightarrow \infty} a$

Example: $A = (0, 2], (a_n)_{n \in \mathbb{N}} = (1/n)_{n \in \mathbb{N}}$ then $a_n \xrightarrow{n \rightarrow \infty} 0 \notin (0, 2] \implies A$ is not closed

Definition 2.4. Compact Set $A \subset \mathbb{R}$ is called compact if for all sequences $(a_n)_{n \in \mathbb{N}}$ with $a_n \in A \forall n \in \mathbb{N}$ for all $n \in \mathbb{N}$ there is a convergent subsequence $(a_{n_k})_{k \in \mathbb{N}}$. A.k.a It's impossible to leave the set from inside the set.

Compact $\implies$ Closed

Examples:

$\emptyset, \{5\}$ is compact

$\mathbb{R}$ is not compact because the points do not accumulate at a specific point

$[c, d]$ with $c \leq d$ is a compact set.

Let $(a_n)_{n \in \mathbb{N}} \subset [c, d] \implies (a_n)_{n \in \mathbb{N}}$ is bounded

Apply Bolzano-Weierstrauss Theorem, we know bd sequences have an accumulation value ($a \in \mathbb{R}$)

We know this accumulation value is $\in [c, d]$ because it's closed so no accumulation value can leave the set.

Definition 2.5. Heine-Borel Theorem For $A \subset \mathbb{R}$, we have:

A is compact $\iff A$ is bd and closed

Proof: ($\leftarrow$) Same as before with Bolzano Weierstrass theorem.

($\rightarrow$) Assume A is compact.

Let $(a_n)_{n \in \mathbb{N}} \subset A$ be a convergent sequence with limit $a \in \mathbb{R}$

Since A is compact, there is a accumulation value, and since it's convergent, there can only be 1 accumulation value therefore there is only 1 possible value a so it is closed.

Assume A is not bounded.

There is a sequence $(a_n)_{n \in \mathbb{N}} \subset A$ with $|a_n| > n \forall n \in \mathbb{N}$. In this sequence, there is no accumulation value therefore it cannot be compact.

3 Series

Definition 3.1. Series Let $(a_n)_{n \in \mathbb{N}}$ be a sequence. The sequence $(s_n)_{n \in \mathbb{N}}$ given by $s_n := \sum_{k=1}^n a_k$ is called a series.

If $(s_n)_{n \in \mathbb{N}}$ is convergent, we write:

$$\sum_{k=1}^{\infty} (a_k) := \lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \sum_{k=1}^n (a_k)$$

Properties:

If $\sum_{k=1}^{\infty} (a_k)$ and $\sum_{k=1}^{\infty} (b_k)$ are convergent, then:

- $\sum_{k=1}^{\infty} (a_k + b_k)$ is convergent
- $\sum_{k=1}^{\infty} (\lambda * a_k)$ is convergent

Definition 3.2. Cauchy Criterion $\sum_{k=1}^{\infty} (a_k)$ is convergent $\iff \forall \epsilon > 0 \exists N \in \mathbb{N} \forall n \geq m \geq N : |\sum_{k=m}^n (a_k)| < \epsilon$

Example: $\sum_{k=1}^{\infty} (-1)^k$ Calculate $|\sum_{k=N}^{N+1} (-1)^k| = |1 - 1 + 1| = 1$. Pick an epsilon less than 1 and Q.E.D.

Observation 3.3. $\sum_{k=1}^{\infty} a_k$ is convergent $\implies (a_k)_{k \in \mathbb{N}}$ convergent with $\lim_{k \rightarrow \infty} a_k = 0$

Definition 3.4. Leibniz Criterion Let $(a_k)_{k \in \mathbb{N}}$ be convergent with $\lim_{k \rightarrow \infty} a_k = 0$ and monotonically decreasing.

Then: $\sum_{k=1}^{\infty} (-1)^k a_k$ is convergent

Now how do we multiply two series together? $\sum_{k=1}^{\infty} (-1)^k a_k * \sum_{k=1}^{\infty} b_k \rightarrow \sum_{k=1}^{\infty} c_k$

Simple Example:

$$(a_0 + a_1 + a_2) * (b_0 + b_1 + b_2)$$

$$= a_0b_0 + a_1b_0 + a_2b_0 + a_0b_1 + a_1b_1 + a_2b_1 + a_0b_2 + a_1b_2 + a_2b_2$$

$$= (a_0b_0) + (a_1b_0 + a_0b_1) + (a_2b_0 + a_1b_1 + a_0b_2) + (a_2b_1 + a_1b_2) + (a_2b_2)$$

We made little packets where the sum of the indices add up to the same number 0, 1, 2, 3, 4, ...

If we made this infinite, the table would keep on growing in size. Packet 1 has 1 element, packet 2 has 2 elements and so on.

Definition 3.5. Cauchy Product For two series $\sum_{k=1}^{\infty} (a_k), \sum_{k=1}^{\infty} (b_k)$, the series $\sum_{k=1}^{\infty} (c_k)$ with $c_k = \sum_{l=1}^k a_l b_{k-l}$ is called the Cauchy product

Observation 3.6. If $\sum_{k=1}^{\infty} (a_k)$ is absolutely convergent and $\sum_{k=1}^{\infty} (b_k)$ convergent then the cauchy product $\sum_{k=1}^{\infty} (c_k)$ is absolutely convergent and $\sum_{k=1}^{\infty} (c_k) = \sum_{k=1}^{\infty} (a_k) * \sum_{k=1}^{\infty} (b_k)$

4 Differentiation

Definition 4.1. Differential Quotient/Derivative $I \subset \mathbb{R}$ interval with more than one point or $I \subset \mathbb{R}$ open set, $f : I \rightarrow \mathbb{R}, x_0 \in I$

We call f differentiable at x_0 if there is a function $\Delta_{f_1 x_0} I \rightarrow \mathbb{R}$ with $f(x) = f(x_0) + (x - x_0) * \Delta_{f_1 x_0}(x)$

$$f'[x_0] := \frac{f(x) - f(x_0)}{x - x_0} =: \frac{df}{dx}(x_0)$$